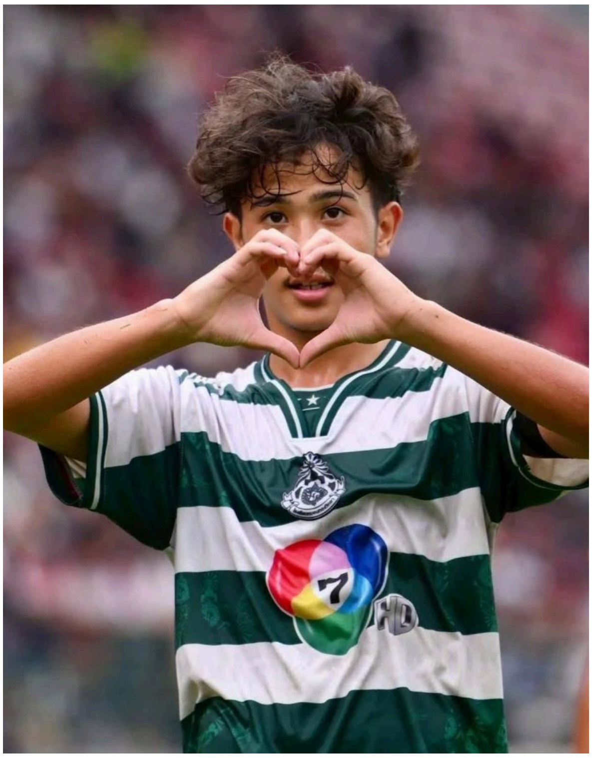
Course: Individual Software Development Process' 2026

Iteration Report: Iteration 1

By TripleT

Chosen Irl Challenge: Project A: Lab Data Management



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Tanisorn Pisittanaphat	6810545671
Tanawat Rungwallapa	6810545646
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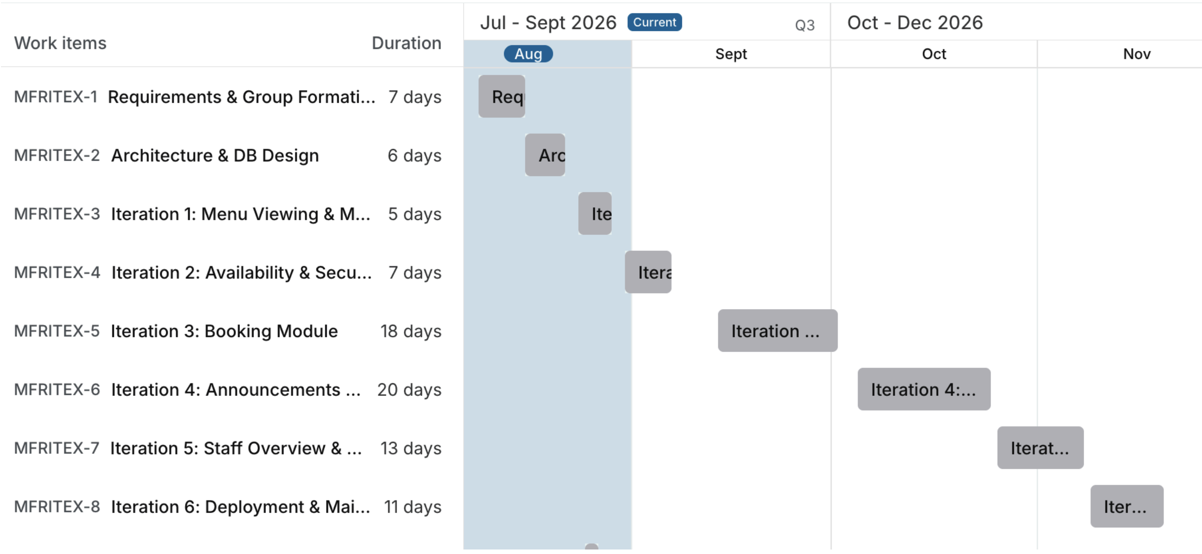
1. Project Snapshot

Note: This section must include both Gantt Chart for the whole project and the calculation of the critical path for the project, and each iteration. Created via Plane.so.

Note: Project Repository is mandatory for the report submission of Iteration 2. For Iteration 1, this can leave blank as "(WIP)".

Project Repository: WIP

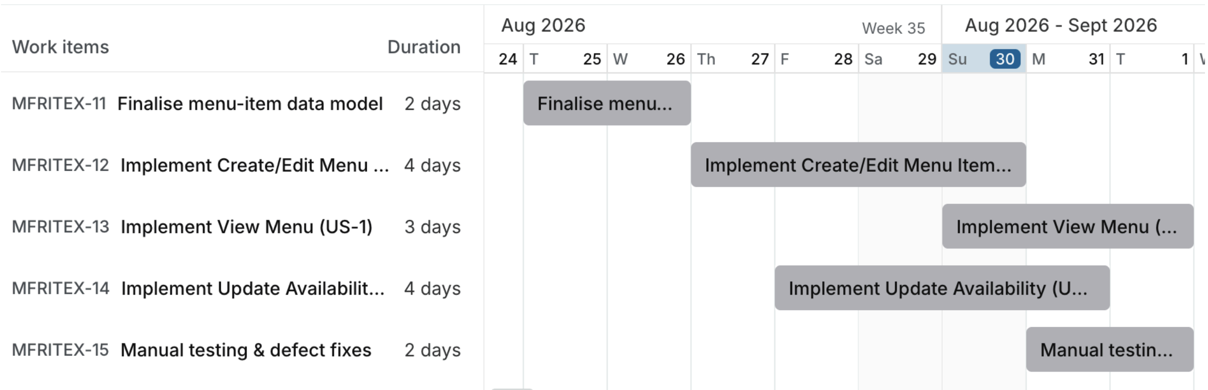
1.1 Project-Wide Gantt Chart



Critical Path: Requirements sign-off → Architecture/DB schema finalisation → Menu Create/Edit implementation (US-2) → Availability update logic (US-3) → Booking module → Deployment

Critical Path Duration: 98 calendar days

1.2 Current Iteration Gantt Chart



Critical Path: Finalise menu-item data model → Implement Create/Edit Menu Item (US-2) → Implement View Menu (US-1)

Critical Path Duration: 7 calendar days

2. Sprint/Board Status Snapshot

Note: User stories strictly reflect the Lab Data Management SRS requirements. Maximum capacity for Iteration 1 is 3 User Stories.
Iteration Goal: Deliver a working, doable resource-viewing and administrative resource record management slice so lab members can view accurate, current lab resources and administrators can create resource records, directly supporting Sub-Goals SG-1 and SG-3 from the SRS.

Capacity Declared: 3 user stories (at the Iteration 1 hard limit; fixed ceiling for Iteration 1).

User Story	Description	Owner	Status	Within Capacity (Y/N)	Notes
US-1	As a lab member, I want to view lab resources so that I can know what resources are available for use.	Pensiri Yakongko	Done	Y	Core browsing grid; connected with backend REST API.
US-2	As a lab member, I want to find lab resources so that I can quickly narrow and organise the resource list.	Tanisorn Pisittanaphat	Done	Y	Implemented search by name and filtering by category; sorting deferred to Iteration 2.
US-3	As a lab member, I want to view detailed information about a resource so that I can understand its specifications, location, and operational status.	Archan Norsiri	Done	Y	Implemented resource detail modal and backend query logic; completed on schedule.

3. Scope Addition/Change Log

Note: Log any item added, expanded, or swapped in after planning began. Empty is a good sign; frequent entries signal scope creep.

Item added/changed	**Reason**	**What as dropped to compensate (one-in-one-out)**
US-2 scope adjusted is Sorting functionality deferred to Iteration 2	IO prioritise completion of the core search and filtering functionality within Iteration 1	Sorting by name, category and availability status was deferred to Iteration 2

4. Retrospective

What went well	**What didn't go well**	**Action items for next sprint**
Team successfully implemented the core functionally for all three planned user stories within the Iteration 1 capacity US-3 was completed ahead of schedule	US-2 could not meet all originally planned acceptance criteria because sorting functionality was deferred to Iteration 2	Complete the deferred sorting functionality early in Iteration 2 and allocate sufficient time for all acceptance criteria
Core resource browsing flow was successfully established, covering resource listing, resource search / filter and resource details	US-1 depended on seed data being prepared before the resource list could be properly implemented and tested	Identify implementation dependencies and prepare required seed and test data before development begins in future iterations

5. Testing & Quality Notes

Note: any bugs, defects, tests (informal & formal) encountered and/or fixed during this iteration can be included here. If there is none, just put "(None applicable)" below.

Manual testing was performed across all three Iteration 1 user stories to verify their acceptance criteria before the demonstration. Resource listing (US-1), Resource search and filtering (US-2) and Resource detail viewing (US-3) were tested during the iteration. Sorting functionality in US-2 was not included in the completed Iteration 1 scope and was deferred to Iteration 2 for further implementation and testing

6. Project Risk Register

Risk ID	**Description**	**Likelihood (L/M/H)**	**Impact (L/M/H)**	**Mitigation/ Contingency**	**Status**

R-01	Incorrect or inconsistent lab resource data is entered into the system.	M	H	Use input validation, required fields, and database constraints. Correct invalid records when identified.	Open
R-02	Two users may attempt to book the same resource for overlapping time periods.	M	H	Check resource availability and existing approved bookings before accepting a booking request.	Open
R-03	Unauthorized users may access administrative functions or modify lab data.	M	H	Use Google OAuth 2.0 and role-based access control. Restrict administrative functions to authorized administrators.	Open
R-04	Incorrect roles or permissions may be assigned to users.	M	H	Restrict role management to authorized administrators and record role changes in the audit log. Revoke or correct incorrect roles immediately.	Open
R-05	Important changes to lab records may not be properly traceable.	L	H	Record the acting user, timestamp, action, and affected record in the audit log for important changes.	Open
R-06	Lab data may be lost or corrupted because of database or application failure.	L	H	Use database transactions, constraints, regular backups, and recovery procedures.	Open

R-07	Booking or issue-report notifications may fail to reach users.	M	M	Ensure status changes are stored in the database and test notification functions. Users can check the current status directly in the system.	Open
R-08	Google OAuth 2.0 authentication may fail or be incorrectly configured.	M	H	Test authentication thoroughly and implement error handling for failed sign-in attempts.	Open
R-09	Integration problems may occur between React, Node.js, and MySQL.	M	M	Define clear API interfaces and perform integration testing regularly during development.	Open
R-10	Team members may have inconsistent development environments.	M	M	Use Docker to provide a consistent development and testing environment across the team.	Open
R-11	Development tasks may not be completed according to the project schedule.	M	H	Divide development into milestones, assign responsibilities, and review team progress regularly.	Open
R-12	Git conflicts or incorrect code changes may affect system development.	M	M	Use GitHub branches, commits, reviews, and regular synchronization. Revert problematic changes when necessary.	Open

7. Individual Contribution Log

Note: One row per member, filled in by that member. The submission of this file considered that the form below is consented and confirmed by the whole team. This section directly supports the Individual Project Contribution and Peer Evaluation grading components.
Note 2: For Iteration 1, evidence can be a reference to the video presentation/iteration report that is related to the responsible task(s) (e.g. Lalita's Evidence). However, for Iteration 2, the reference must be on **GitHub** file and commit only, which is uploaded by that team member.

Student Name	**Tasks owned this sprint**	**Course concept(s) applied**	**Evidence**	**Self-rated effort (1-5)**
Archan Norsiri	US-3 (View Resource Details) Implementation resource detail data handling	Applied requirement traceability and data mapping to ensure the displayed resource information matched the SRS and acceptance criteria	-	5
Pensiri Yakongko	US-1 (View Lab Resources) Frontend and resource-list display	Applied requirement to interface traceability to present current lab-resource information according to the defined user story and SRS	-	4
Tanisorn Pisittanaphat	US-2 (Find Lab Resources) Search and filtering functionality	Applied search and filtering requirements and user interface logic to help users narrow the lab resource list	-	4
Tanawat Rungwallapa	Manual testing across US-1, US-2 and US-3; Acceptance criteria checking and defect logging	Applied informal software testing concepts to verify the implemented functionality against	-	4

		acceptance criteria before the demo		
Panuwitch Sowkasem	Project and Iteration 1 Gantt charts critical path calculation	Applied project management fundamentals, including scheduling and the Critical Path Method	-	3